Presentation Report: The Large Hadron Collider and Flavor Anomalies

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1 Introduction

In this report, we trace the path of the proton beam briefly, starting with its production and all the way to LHCb. I will summarise recent searches for New Physics using B meson decays at LHCb. There was a lot of cautious excitement in the Physics community due to discrepancies(compared to what's predicted by the Standard Model) in P_5' value in B meson decays into muons along with anomalies in the Branching Ratio of electrons and muons. We'll see why there is a doubt about whether this suspicion is true or not. I'll end with our current standing in the context of all the observations and analyses at the end of Run 2 of the LHC.

2 LHC: Bird's Eye View

LHC, as it meets the common eye, is a circular tunnel, 27 km in circumference, 100 m underground. The ring geometry is divided into exactly 8 points, 4 of them being the detector points- ATLAS, CMS, ALICE and **LHCb**, and 4 being utility points dedicated to RF acceleration, beam cleaning, and safe dumping. Protons run in two parallel beam pipes in opposite directions.

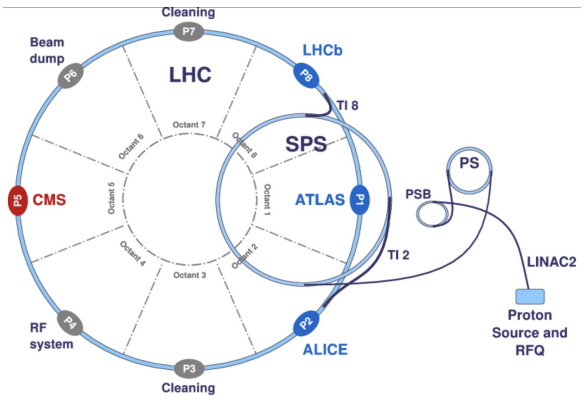


Figure 1: Schematic layout of the LHC ring and its detectors.

2.1 The Proton Injection Chain

We first need to understand at least briefly how the protons, the collisions of which give us hints to so much new physics, are produced, accelerated, sent into the main 27km ring in LHC and then how we maintain them to appropriate dimensions for the collisions to occur.

2.1.1 Source

Everything starts with a canister of Hydrogen gas put inside a Duoplasmatron, inside which we put enough electric field to break apart the protons and electrons. If we look at both the pipes, there are almost $\mathbf{6 \times 10^{14}}$ of them. Though this seems like a large number, even a cubic

centimeter of hydrogen gas consists around $\mathbf{5 \times 10^{19}}$ of hydrogen atoms. The protons are then accelerated with a voltage of **90kV** and reach to **1.6** percent of the speed of light before leaving the duoplasmatron. They're further accelerated by radio frequency quadrupole and focussed by it to make the beam narrow. This is also the place where protons are first bunched so the beam is not continuous but consists of protons in groups. They're then sent to the first of the four accelerators- LINAC2(upgraded to LINAC 4 which takes in $\mathbf{H}^-$ rather than protons.LINAC4 uses a RF Volume ion source instead.)

2.1.2 LINAC2

LINAC2 is a linear accelerator, which accelerates the protons by providing an oscillating electromagnetic field by creating RF standing waves in a copper cavity. However, because this field oscillates, though it accelerates the protons when pointing forward, the effect is reversed when the direction of the field reverses. Therefore, a proton sitting in the cavity will gain zero net acceleration. To fix this, we have drift tubes in the cavity, which are made out of metal cylinders acting as a Faraday cage, shielding the protons from the backward point electric field. The protons "drift" at a constant speed inside these tubes. Each time the protons are accelerated through EM fields, they gain more speed and so these tubes keep on increasing in length along the length of the accelerator. LINAC 2 accelerates the protons upto an energy of 50 MeV.

As I mentioned, LINAC 4 replaced LINAC 2 (in 2020). There are two major changes: instead of protons, LINAC 4 accelerated $\mathbf{H}^-$ ions, and the frequency of the electromagnetic field. Liouville's theorem puts a restriction on the density of protons in the beam. By using $\mathbf{H}^-$ ions and then later passing them through a carbon strippig foil to strip off the electrons to merge them with the already existing proton beam, LINAC4 bypasses Liouville's theorem, increasing beam's brightness. It also runs at a higher frequency(352 MHz contrary to LINAC2's 200 MHz) and uses a staged sequence of different accelerating structures, unlike just one in LINAC 2. This sequence allows LINAC 4 to accelerate the protons upto 160 MeV. This reduces the space charge repulsion.

2.1.3 Proton Synchroton Booster

PSB is designed to receive protons from LINAC2/LINAC4 and starts accumulating them. As we're accumulating more and more charges in the same volume, the Coulomb repulsion keeps on increasing, which has the effect of expanding the beam and pushing the particles into pipe walls(this effect is prominent since protons are still not at relatively very high energy). To get over this problem, PSB actually distributes the protons in 4 vertically stacked rings(thus reducing charge density in individual rings), each of them being 157 m in circumference. The PSB's radio cavities now accelerate the protons to significantly higher relativistic speeds(energy upto 1.4 GeV). At these speeds, the Coulomb space charge effect is no more a problem. Then, PSB recombines the protons from 4 rings into a single beam to send it to the next component before LHC main ring- Proton Synchroton.

2.1.4 Proton Synchroton

As already mentioned, proton beam is not exactly continuous but consists of many protons bunched together. I also have mentioned that this is done with RF cavities. Consider a reference proton. If we want a bunch of protons to stay together while acceleration, any proton arriving before this reference proton will arrive at the time when the voltage is climbing up and still below what the reference proton will see it to be, and so it gets a lower forward kick and the opposite happens with a proton that arrives late- it sees a higher voltage than the reference proton and gets a more forward kick. PS is exactly 628 m in circumference(recall that PSB had each ring's circumference as 157 m and 628m is just four times that as required).The PS

has two objectives: to handle the relativistic transition energy and to create the space between bunches as required by LHC.

- Phase jump: At lower energy, in the classical realm, a proton with more energy moves faster and completes its orbit in a lower time. To maintain bunching, the RF cavities are timed so that so that a particle arriving early gets less of a forward kick and a particle arriving relatively late gets more of a forward kick. However, at 6.1 GeV in PS, the protons cross a relativistic threshold known as the transition energy(γ_t). At γ_t, the protons are moving so close to the speed of light that adding more energy does not increase their velocity, rather forces them into a wider orbital radius. Now, such a proton takes more time to complete an orbit. If RF cavities maintained their original timing, the bunches would tear apart. To get over this problem, as soon as the beam crosses 6.1 Gev, the RF system flips the phase, restoring the beam stability.

- Bunch splitting: The LHC requires a strict 25ns spacing between proton bunches to operate detectors. PS, instead of operating the RF cavity at a single frequency, uses tunable cavities operating at multiple harmonics(10MHz, 20MHz, 40MHz, 80MHz). This slices a single, long proton bunch received from PSB into two or three shorter bunches. Through a sequence of triple and double splittings, the PS divides the entire beam into a train of 72 bunches spaced by exactly 25ns.

2.1.5 Super Proton Synchroton

SPS receieves bunched 26 GeV beam from PS and accelerates them upto 450 GeV for final injection into LHC.

3 Orchestrating the Collisions

- **Maintaining the beam:** When the beam is injected from SPS into the main ring, it has tiny momentum in the transverse direction. Even if a proton is very very slightly misaligned, say, just 1 microradian, it will drift 2.7 centimeters off center in just a single lap. The LHC pipe is just about 5 cm wide and so by the second lap, the proton will crash into the wall. We need a focusing mechanism to fix this. We use magnetic quadrupoles to focus the beam. A magnetic quadrupole develops a restoring force in one direction but has an effect in the other transverse direction. Therefore, we use them in series, at 90 degrees relative to each other, focusing in one direction and then defocusing in that very direction. Because the magnetic force is proportional to the distance from the center of beam propagation, defocusing is not as strong as focusing, so the net effect is focussing in both the directions. Such a setting is called a FODO(Focus-drift spaces(O)-defocus-O) cell. This maintains beam width to 1 mm. This is also very useful because to have a good number of collisions, we need the beams in the two pipes to be very narrow. The FODO lattice maintains beam to 1mm which later is further narrowed down to 16 μm using inner triplet quadrupoles.

- **Merging the beams:** For the majority of the 27-kilometer LHC ring, the two counter-rotating proton beams travel isolated in separate, parallel vacuum pipes, which are encased together within the 'Twin-Bore' superconducting dipole magnets. To make collisions happen at the experimental interaction points, approximately 130 meters before a detector, specialized separation and recombination dipole magnets (D2 and D1) steer the two independent beams inward, physically merging their paths into a single, shared vacuum chamber.Despite sharing this final 130-meter stretch of beam pipe, the proton bunches do not collide prematurely because they are not steered perfectly head-on. Instead, the

magnetic optics direct the beams to intersect at a crossing angle of approximately 285 microradians. This precise 'X' like formation ensures that the bunches from the two beams only pass through each other at the exact geometric center of the detector. Immediately prior to this intersection, the Inner Triplet quadrupole magnets violently squeeze the beams down to a 16 μm width, maximizing the collision rate (luminosity) exactly at the interaction point.

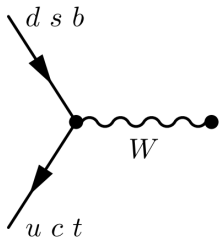
4 B Decays and Anomalies

Flavour physics divides Lepton Flavour Universality tests into two distinct sectors based on their Feynman topologies: allowed tree-level transitions and suppressed quantum loops.

4.1 Charged and Neutral Currents

4.1.1 Charged currents:

The W bosons couple up type quarks(up, charm, top) to down type quarks(down, strange, bottom).



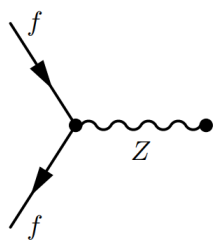
$$\mathcal{L}_{\text{CC}}^{\text{quarks}} = -\frac{g}{\sqrt{2}} \sum_{i,j=1}^{3} \bar{u}_i \gamma^\mu P_L V_{ij}^{\text{CKM}} d_j W_\mu^+ + \text{h.c.} \tag{1}$$

Tree-level flavour transitions(like b $\rightarrow$ c) are allowed.

4.1.2 Neutral currents:

The Z boson couples same fields to themselves. Because the CKM matrix is unitary, off diagonal flavour mixing terms cancel out.

$$\mathcal{L}_{NC} \supset \frac{g}{2\cos\theta_W} Z_\mu \left(\bar{b}_L \gamma^\mu b_L + \bar{s}_L \gamma^\mu s_L\right) \tag{2}$$



Tree level flavour transitions(like $b \rightarrow s$) are mathematically forbidden. They must proceed via loops.

4.2 Allowed vs. Suppressed B Decays

In the Standard Model, B meson decays provide a stringent test of Lepton Flavour Universality (LFU). The observables can be broadly categorized by their underlying weak interaction mechanisms:

A. Charged Current: $b \to c\ell\nu$

Tree-Level Allowed → Large decay rates

These "semileptonic decays" are mediated by a $W^{\pm}$ boson at tree level. The relevant LFU observables are defined as the ratios of the heavy tau branching fraction to the lighter muon branching fraction:

$$R_D = \frac{\mathcal{B}(B \to D\tau\nu)}{\mathcal{B}(B \to D\mu\nu)} \tag{3}$$

$$R_{D^*} = \frac{\mathcal{B}(B \to D^*\tau\nu)}{\mathcal{B}(B \to D^*\mu\nu)} \tag{4}$$

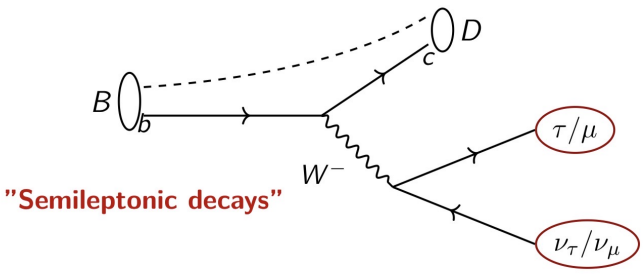


Figure 2: Tree level allowed B decays.

B. Neutral Current: $b \to s\ell^+\ell^-$

Loop Suppressed → Rare decays (FCNC)

Because the Z boson strictly forbids flavour mixing, these "rare decays" must be mediated by electroweak penguins or box diagrams. The observables compare muon to electron channels:

$$R_K = \frac{\mathcal{B}(B^+ \to K^+\mu^+\mu^-)}{\mathcal{B}(B^+ \to K^+e^+e^-)} \tag{5}$$

$$R_{K^*} = \frac{\mathcal{B}(B^0 \to K^{*0}\mu^+\mu^-)}{\mathcal{B}(B^0 \to K^{*0}e^+e^-)} \tag{6}$$

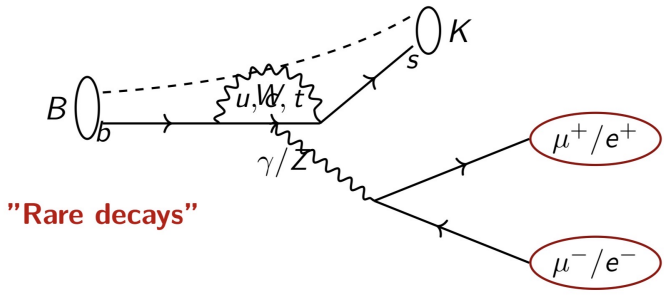


Figure 3: Loop-suppressed rare B decay via electroweak penguin.

I want to point out the reason for using ratios of branching fractions: Calculating absolute branching ratios $\mathbf{B}(B \to D * l\nu)$ requires knowing hadronic form factors which bring with themselves uncertainties from Lattice QCD. By taking the ratio like $\mathbf{R_{D*}}$, these theoretical uncertainties cancel out, resulting in a more precise standard model prediction.

4.3 Anomaly in the charged current sector: R_{D^*} and R_D

As already mentioned, charged current sector is probed via the R_D and R_{D^*} ratios, which compare the branching fraction of $B \to D^{(*)}\tau\nu$ to the light lepton channels (e, μ). In the Standard Model, the W bosons couple to e, μ , τ with the exact same gauge coupling. The only reason $R_{D^*} \neq 1$ in the SM is strictly due to the fact that τ mass is much larger than μ mass. The HFLAV 2023 plot in figure 4 shows the difference in SM prediction and world observational average:

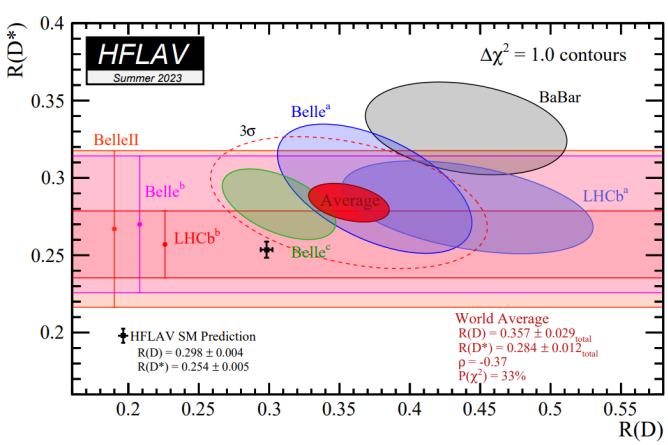


Figure 4: Experimental status(HFLAV 2023)

Current world averages reflect a combined $\sim 3.2\sigma$ tension, showing an excess of taus. Explaining a tree-level anomaly requires strong or light New Physics (e.g., a $1-2$ TeV Leptoquark) coupling preferentially to the third generation.

4.4 Observations in Rare Decay sector

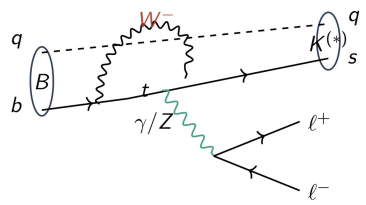


Figure 5: FCNC loop diagram

To theoretically describe B meson decays, we use the Operator Product Expansion (OPE). We integrate out the heavy, short-distance degrees of freedom (the W and Z bosons, and the top quark) to get the full Standard Model action reduced to a low-energy Effective Field Theory (EFT).

For $b \to s$ flavour-changing neutral current (FCNC) transitions, the Weak Effective Hamiltonian is factorized as:

$$\mathcal{H}_{\text{eff}} = -\frac{4G_F}{\sqrt{2}} V_{tb} V_{ts}^* \frac{e^2}{16\pi^2} \sum_i \left(C_i(\mu) O_i(\mu) + C_i'(\mu) O_i'(\mu) \right) + \text{h.c.} \tag{7}$$

where G_F is the Fermi coupling constant and $V_{tb}V_{ts}^*$ are the Cabibbo-Kobayashi-Maskawa (CKM) matrix elements for a specific flavour transition.

In our context, it is enough to understand that any New Physics does not change the structure of the low energy operators(O_i, but shifts the Wilson coefficients: $C_i = C_i^{\text{SM}} + C_i^{\text{NP}}$.

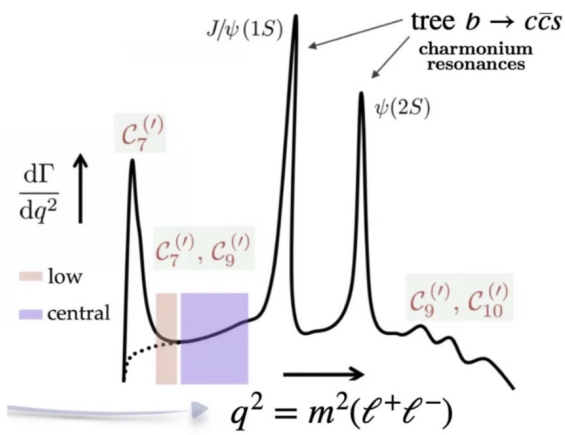


Figure 6: Differential decay rates

The q^2 bins we want to look into: In the Feynman diagram, q is the four momentum of the virtual particle. q^2 is also the invariant mass of the muon pair. Figure 6 shows how the decay rate changes as a function of q^2. q^2 ranges from $0.045GeV^2$ to $22.9GeV^2$. The resonant peaks correspond to regions where q^2 equals the mass squared values of the charm mesons. When we do observations, we want to ignore these resonant regions because they happen at tree level(large decay rates) and blind us to the loop level rare decay we're interested in.

Following are the key observables that we will survey to look for New Physics:

- **Differential decay rates**: Contain large hadronic(form factors) theoretical uncertainties($20 - 30\%$).
- **Angular** $P5'$: Cancel leading form factor dependencies.
- **Universality**(R_K, R_{K*}): Theoretically most precise prediction.

4.4.1 Differential decay rates of Rare Decays: $b \to s\mu_+\mu_-$:

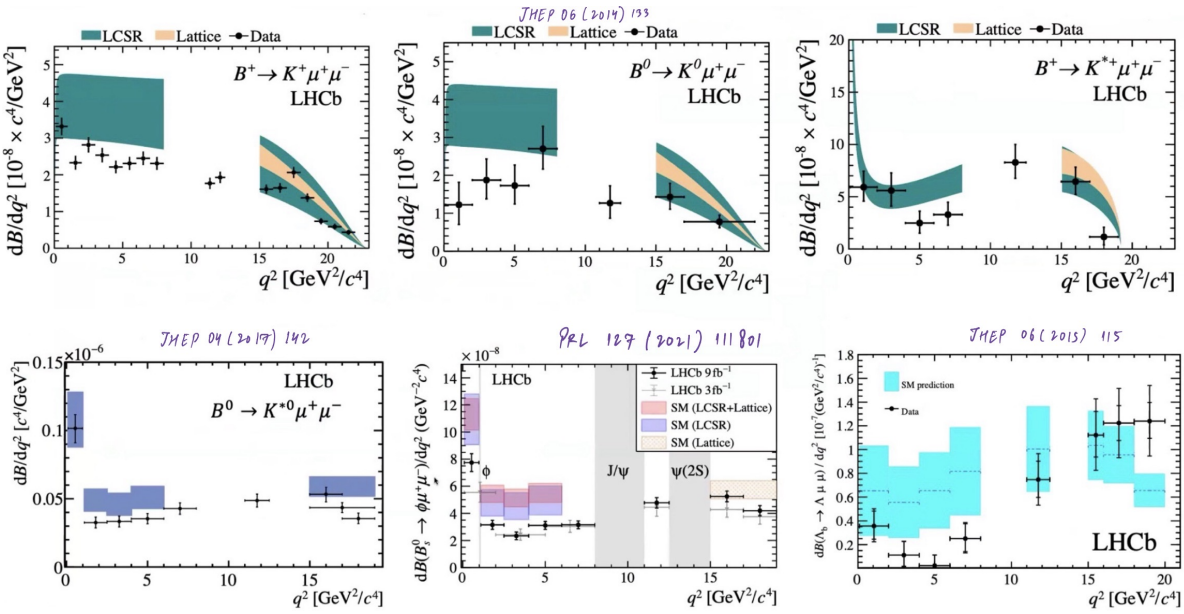


Figure 7: Differential decay rates

Above are the plots of branching ratios with respect to q^2. There are a few things to note:

- 1. The green and orange shaded area are so wide because of the uncertainties in Lattice QCD and Light Cone Sum Rules, which are the techniques we use to do non perturbative QCD calculations.
- 2. Branching fractions are plotted for different Hadronic environments (different spectator quarks involved).
- 3. The branching fractions are consistently lower than the Standard Model predictions.

The question is: Is it a true anomaly or just theory uncertainty? This is not clear at the level of differential decay rates.

4.4.2 The Optimized angular P_5' Anomaly in $B^0 \to K^{*0}\mu^+\mu^-$ Decays

A. Angular distribution

The rare, loop-suppressed neutral current decay $B^0 \to K^{*0}\mu^+\mu^-$ results in a four-body final state, as the vector meson K^{*0} subsequently decays strongly via $K^{*0} \to K^+\pi^-$. This final state is defined by three angles:

- θ_K: The angle of the Kaon (K^+) in the K^{*0} rest frame.
- θ_l: The angle of the anti-muon (μ^+) in the dimuon rest frame.
- ϕ: The angle between the hadronic plane (defined by the $K\pi$ system) and the leptonic plane (defined by the $\mu\mu$ system).

The full differential decay rate is expanded in terms of spherical harmonics dependent on these three angles, in a sum parameterized by angular coefficients, J_i:

$$\begin{aligned}\frac{d^4\Gamma}{dq^2\, d\cos\theta_K\, d\cos\theta_l\, d\phi} = \frac{9}{32\pi}\Big[& J_{1s}\sin^2\theta_K + J_{1c}\cos^2\theta_K \\ & + (J_{2s}\sin^2\theta_K + J_{2c}\cos^2\theta_K)\cos 2\theta_l \\ & + J_3 \sin^2\theta_K \sin^2\theta_l \cos 2\phi \\ & + J_4 \sin 2\theta_K \sin 2\theta_l \cos\phi \\ & + J_5 \sin 2\theta_K \sin\theta_l \cos\phi \\ & + (J_{6s}\sin^2\theta_K + J_{6c}\cos^2\theta_K)\cos\theta_l \\ & + J_7 \sin 2\theta_K \sin\theta_l \sin\phi \\ & + J_8 \sin 2\theta_K \sin 2\theta_l \sin\phi \\ & + J_9 \sin^2\theta_K \sin^2\theta_l \sin 2\phi \Big]\end{aligned} \tag{8}$$

J_i coefficients depend on transversity amplitudes, which then depend on the Wilson coefficients (e.g., C_7, C_9, C_{10}) and the hadronic form factors.

B. Uncertainties Optimized Observables

Theoretical predictions for the required hadronic form factors, calculated via Lattice QCD at high q^2 and Light-Cone Sum Rules (LCSR) at low q^2, carry uncertainties.

We can construct an observable $P5'$ which is constructed from taking such a ratio which cancels these form factor uncertainties:

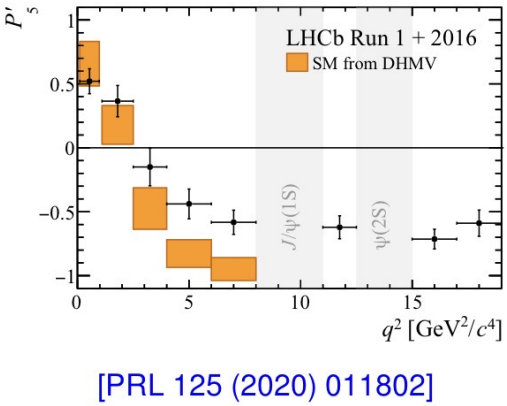
$$P_5' = \frac{J_5}{2\sqrt{-J_{2s}J_{2c}}} \tag{9}$$

Now to get J_5 experimentally, we use a folded asymmetry trick to extract out J_5. Look at the plot alongside. The LHCb counts the events in the blue bins and subtract the events that fall in the red bins. This is equivalent to integrating the entire 11 term equation(9) over all angles multiplied by a sign function $\mathbf{sgn}(cos\theta_k cos\phi)$. Only the J_5 term survives.

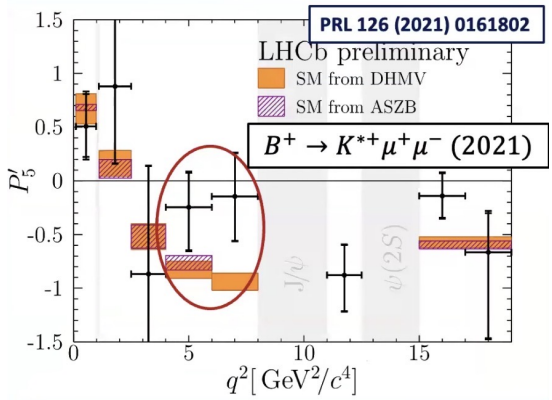
Now, J_5 is very sensitive to C_9 so if we see an anomaly in $P5'$, it must be interpreted in terms of C_9 which is the vector coupling coefficient.($O_9 = (\bar{s}\gamma_\mu P_L b)(\bar{\ell}\gamma^\mu \ell)$)

C. Data

Anomaly: Measurements of P_5' by the LHCb collaboration have consistently shown a deviation of approximately 2.5σ to 3σ from the Standard Model prediction, in the intermediate $4 < q^2 < 8$ GeV^2 bins.

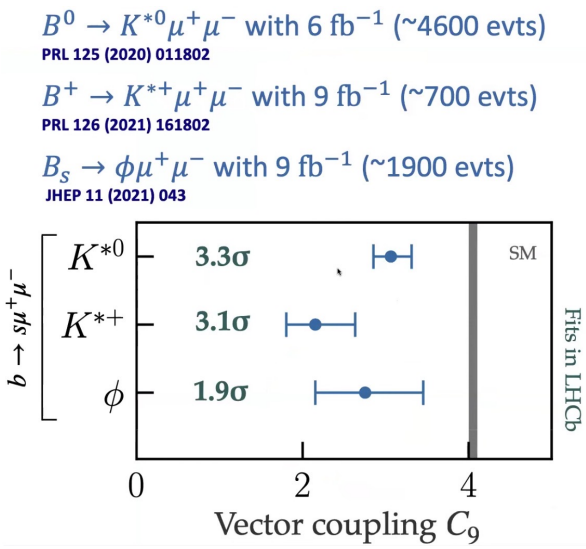


(a) $B^0 \to K^{*0}\mu^+\mu^-$



(b) $B^+ \to K^{*+}\mu^+\mu^-$

The interpretation of P_5' discrepancy can be interpreted in terms of a shift required in C_9 vector coupling coefficient.(as shown in figure 8)



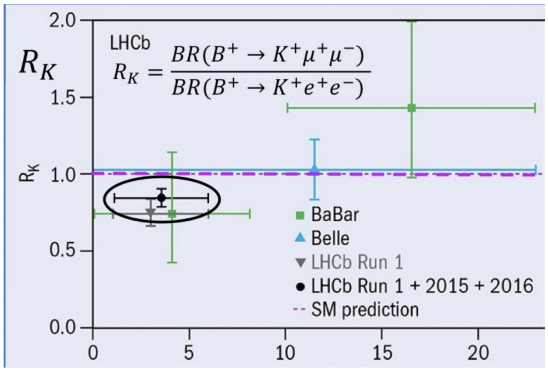
The C_9 deficit: Deviation from standard model prediction is by $\Delta C_9 \simeq 1$. An important thing to note in figure 8 is that no matter what the spectator quark is, the coupling shifts the same. This strongly disfavours localised QCD form-factor errors. Also, since the shift is in the vector coupling Wilson coefficient, what this suggests is that if all this data is pointing towards New Physics, it is parity conserving.

4.4.3 Universality: R_K and R_{K^*}

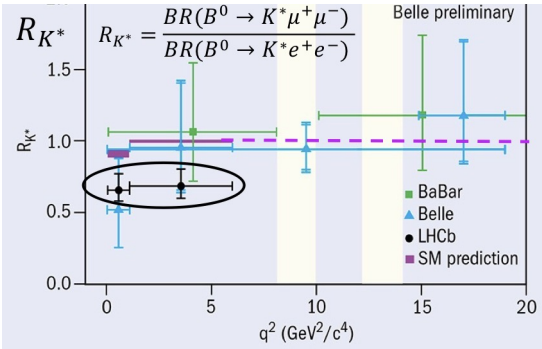
$$R_K = \frac{\mathcal{B}(B^+ \to K^+\mu^+\mu^-)}{\mathcal{B}(B^+ \to K^+e^+e^-)} \tag{10}$$

$$R_{K^*} = \frac{\mathcal{B}(B^0 \to K^{*0}\mu^+\mu^-)}{\mathcal{B}(B^0 \to K^{*0}e^+e^-)} \tag{11}$$

The experimental data for the ratios are demonstrated in the plots below.



(a) figure 9.a



(b) figure 9.b

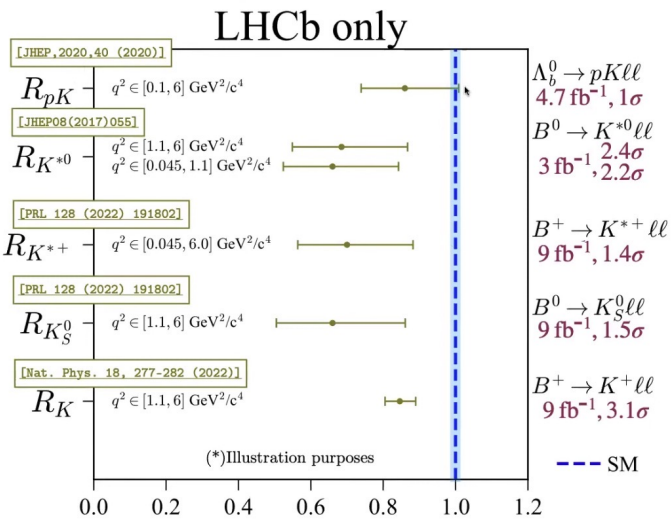


Figure 10

These plots show that the ratio of branching fractions of electron and muon channels is considerably less than 1(deficit in muon production) which is in contradiction with the Standard Model prediction(Lepton Flavour Universality). Also note that this is the most theoretically robust method.

Along with the discrepancies shown in section 5.4.2, this showed something is going on in the muon channel.

5 March and December, 2022

5.1 Anomaly!

We saw that there are anomalies in angular geometry(P_5') in B meson rare decay to muons dictating a shift in C_9. Also, there are anomalies in the branching ratios(R_K, R_{K^*} confirm the same muon deficit at 3σ significance. Recall that these two observables have much more theoretical certainty than differential decay rates which also showed some anomaly. All this data, in **March 2022**, seemed to point strongly towards Heavy New Physics(e.g. Z' bosons or Leptoquarks) coupling preferentially to the second generation.

5.2 Misidentification

It is very hard to do precise Physics involving electrons. This is because electrons while passing through detector materials, emits a large amount of Bremsstrahlung radiation. This was taken care of by experimentalists but there was something else which we did not know was making us overestimate the electron counts. Strict re-analysis done in**December 2022** found that kaons and pions were occasionally faking electron signatures in the detector. This brought down the branching ratios due to overcounting of electrons. When these fake signals were properly subtracted out, the R_K ratios increased:

$$R_K = 0.994 \pm 0.042 \tag{12}$$

$$R_{K^*} = 1.027 \pm 0.072 \tag{13}$$

The result is that the Lepton Flavour Universality holds perfectly within current LHCb precision.

6 Summary

- A. Neutral Currents($b \to sl^+l^-$) :
  LFU Holds: The R_K and R_{K^*} anomalies vanished upon correcting hadronic misidentification in the electron channels.
  The Muon Deficit: The $P_5^{'}$ anomaly($\Delta C_9 \simeq -1$ persists, but confusion exists because a charm loop can mimic the same effect.

- B. Charged Currents($b \to c\tau\nu$):
  τ **excess:** A combined 3σ tension remains in the tree level R_D and R_{D^*} ratios.
  Theoretical tension: Explaining a tree level(which is large in terms of cross section) anomaly requires strong/light New Physics, creating tension with the lack of direct particle discoveries at ATLAS and CMS.

- **C. What's next**: The resolution of the R_K anomaly points to the complexity of hadronic backgrounds.
  Run 3 data from LHCb and independent cross checks from Belle II are required to determine if the R_{D^*} anomalies are genuine New Physics or systematic artifacts.

Project Title : Large Hadron Collider

Name: Pushkar Tripathi-Rohit Das

Project Supervisor: Dr.Balasubramanian Ananthnarayan

SR no.: 26474

Institute: Centre for High Energy Physics, IISC Bangalore

HE 397

1 Introduction

Although the Standard Model very greatly describes different interaction except gravity ,there are some unanswered questions. For eg. approx 20 free parameters, bound state of quarks, axion like particles,state of matter in early universe (quark gluon plasma)and string theory predictions .etc . The LHC is located at CERN: Conseil Européen pour la Recherche Nucléaire (European Organization for Nuclear Research) on the Franco-Swiss border near Geneva. As it Operates on a Large energyhence it's design scale is also Large. It's underground as it's very Large and helps in radiation shielding . Hadrons are preferred as they are charged, stable and have high mass (so less energy decay by synchrotron radiation) . Collider is a machine where counter-circulating beams collide. It's advantage over fixed target accelerators are that a beam of same energy produces a much larger collision energy. The main experiments at LHC are:
(a) ALICE-quark gluon plasma
(b)LHCb-Dark matter
(c) ATLAS -General purpose
(d)CMS-General purpose
Proton proton collisions generate energy of order 14TeV. While Pb having lots of Protons can generate energy around 1150TeV . Protons travel in bunches . Each beam has about 2800 bunches and each bunch has 100 billion protons .

Here we go through a review of the two general purpose detector: ATLAS and CMS .

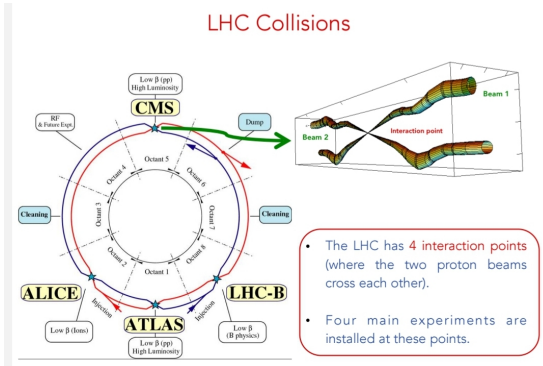


Figure 1: LHC

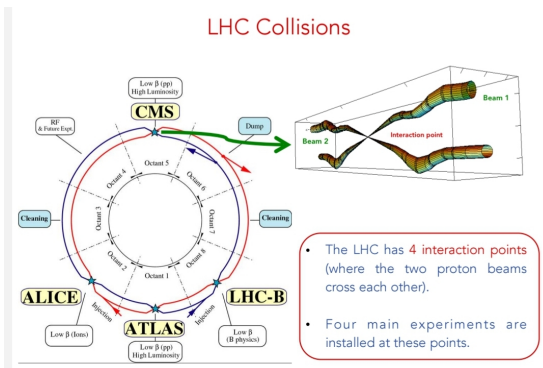


Figure 2: ATLAS run 3 configuration

2 ATLAS

A Toroidal LHC Apparatus (ATLAS) is a general purpose detector used in investigation of Higgs,dark matter , physics in extra dimension. etc . Works in condition created similar to that after big bang .

2.1 ATLAS Run 3 configuration

It uses a right handed coordinate system with origin at the interaction point . It covers almost the entire

solid angle . It contains inner tracking detector surrounded by a thin superconducting solenoid, electromagnetic and hadronic calorimeters, and a muon spectrometer. It incorporates 3 Large air core toroidal superconducting magnets. The dimensions are 25m in height,44m in length and weighs around 7000 ton.

2.1.1 Radiation and shielding

The dominant source of radiation comes from p-p collisions at the interaction point (IP).Each collision produces a cascade of secondary particles that travel through different layers . In the inner tracking detector secondary hadrons dominate at smaller radii while neutrons at larger . Collimators and Liquid Argon calorimeters are used to absorb most of the radiation . Primary particles hit the beam pipe at shallow angles increasing the effective thickness.

2.1.2 Tracking system

It operates inside a 2T axial magnetic field of central solenoid . It contains 3 main subsystems:
(1)Pixel detector: Innermost subsystem of ID. It's used for high spatial resolution and vertex reconstruction . It operates with a 40 MHz bunch crossing frequency and must withstand a high-particle flux and high radiation doses.
(2)Semiconductor Tracker: It's outside Pixel detector and use for precise position tracking.
(3) Transition Radiation Tracker :It's the outermost part

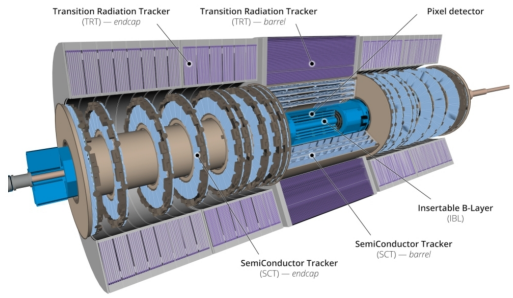


Figure 3: ATLAS ID

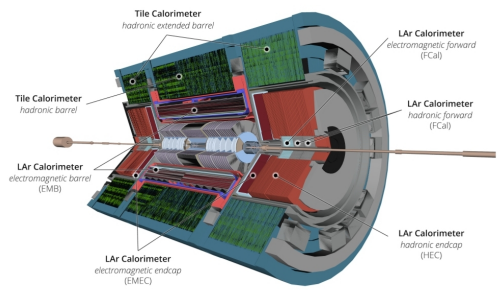


Figure 4: ATLAS Calorimeter

. It detects particles with pseudorapidity ($\eta = -ln[tan(\theta/2)]$) $|\eta| < 5$ and transverse momentum greater than 0.5GeV. It also detects X rays and identifies electrons.

It has 2 regions: Barrel-The detectors are arranged as concentric cylinders and End-cao-the detectors are arranged perpendicular to beam line .Detectors are cylindrical as particles spread out spherically from IP.By measuring the position and hence curvature in magnetic field we calculate momentum.

2.1.3 Calorimeter

It measures total energy of incident particles for detection. It detects both electromagnetically and strongly interacting particles. It uses sampling Calorimeter which has alternate layers of absorber and active detector (measures energy) material. Mainly 2 types of calorimeters are used:Liquid Argon calorimeter(LAr calorimeter):uses liquid argon at -184 °C as the active medium to detect electrons, photons, and some hadrons.
Tile Calorimeter: It surrounds the LAr system and is responsible for measuring the energy of hadrons that penetrate the inner layers.It consists of steel plates as absorbers and plastic scintillating tiles as the active material.It can detect missing energy indicating undetected particles.
It covers $|\eta| < 4.9$. Instead of tracking it completely stops a particle and measures the deposited energy.After hitting dense absorber material it produces secondary particles ,whose energy is detected by active layer .

2.1.4 Muon System

Its the outermost part of detector. As muons are weakly interacting they pass through the Calorimeter. It uses Large air core superconducting toroidal magnets. It consists of a barrel region(3 cylindrical detector layers), endcap region (3 detector disks) and an extended ring detectors .In barrel region $|\eta| < 1.4$, endcap $1.6 < |\eta| < 2.7$ and Transition region 1.4 $|\eta|$ <1.6 . For 3rd run the inner endcap detectors are replaced byNew small

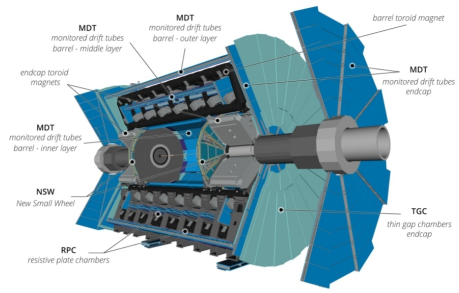


Figure 5: ATLAS Muon System

wheels .Inner endcap receives high flux , NSW increases the effectiveness. Monitored drift tubes(They consist of 30 mm diameter aluminium tubes filled with pressurized gas; a central wire collects ionization as electrons drift toward it.) are used for high precision Tracking and Resistive plate chambers(They provide the Level-1 (L1) trigger and measure the "second coordinate" (non-bending plane) to complement the MDTs.) and thin gap chambers(tracks in non bending plane) inform the system when a muon passes in ns. The latter two are gaseous detectors.

2.1.5 Forward detector

It detects and meadures energy of particles emitted near beamline axis.
(1) Luminosity Cherenkov Integrating detector (LUCID): It measures the luminosity of p-p collisions using Cherenkov radiation.
(2)Absolute Luminosity for ATLAS (ALFA) : It measures total cross section using Scintillator fibres.
(3)Zero degree Calorimeter (ZDC): Detects neutral par-

ticles near beamline axis.
(4)ATLAS Forward Proton detector (AFP): Detects Forward Protons that remain intact after collisions.

2.1.6 Triger and Data Acquisition System

ppp collisions occur at a very high rate 40MHz. It's not feasible to store all events . Hence the system Selects only the interesting event for analysis.
Level 1 trigger is hardware based. It identifies interesting event with a decision time of 2.5 microseconds. It reduces the rate from 40MHz to 100kHz. It takes input information from Calorimeter and muon system.
High Level trigger (HLT) is software based. It performs detailed analysis of events on Large computer farms and stores data for offline analysis. It reduces the rate from 100kHz to 3kHz.
Data Acquisition System(DAQ) manages the flow of data from detector to storage . Read Out system collects data from front end electronics with acceptance rate 100kHz. In run 3 typical event size is around 2.1MB. Total data throughput is around 6.9 GB/s with max capacity of system being 8GB/s .Mass storage capacity is approx. 1.4 PB.

2.2 Experiments at ATLAS

2.2.1 1st run (2009-13)

(a) Recorded $5fb^{-1}$ (integrated luminosity) for p-p collisions at $\sqrt{s}$=7TeV . At $\sqrt{s}$=8TeV it became $21fb^{-1}$
(b) Reached 167 μb^{-1} of Pb-Pb collisions during 2010-11 phase.

(c)Recorded 29.8 fb^{-1} for p-Pb collisions in 2013.
(d)This phase of run was instrumental in Higgs discovery.

2.2.2 2nd run(2015-18)

(a)Reached p-p collisions at $\sqrt{s}$ = 13TeV with luminosity 147 fb^{-1} .
(b) 179.8 nb^{-1} were Recorded for p-Pb collisions.
(c) 1.79 nb^{-1} was Recorded for Pb-Pb collisions.
(d) Highest instantaneous luminosity reached was $2.1 \times 10^{34} cm^{-2} s^{-1}$.
(e)Helped to establish Higgs- fermion interaction and spontaneous symmetry breaking of Electroweak theory.

2.2.3 3rd run (2022-)

(a) In run 3(2022-) the centre of mass energy $\sqrt{s}$ is proposed to increase to 13.6 TeV .
(b)Data(301fb^{-1}) from 2015-24 has led to the first evidence of high transverse momentum Higgs. This is helpful for more sensitive measurement on properties of higgs .
(c)New and stricter constraints on higgsino mass splitting has been achieved. They act as dark matter candidates but are difficult to observe as they mix with other supersymmetric particles
(d)Using luminosity of 164fb^{-1} pp collisions better constraints on CP violating odd parameter and structure of HVV coupling is obtained .
(e)Obtained ttH production signal of 0.63 (SM expected

1.0) which is quite near taking into account the errors. CP mixing angle greater than 62 degree was rejected with confidence 68 percent.
(f)By studying HH to bb$\gamma\gamma$ channel limits have been set on Higgs self interaction.
(g) In run 2 the rare VVZ 3 massive vector boson production was detected. It's very rare due to the high mass involved . Experiments are ongoing to study them with higher precision.

3 CMS

Compact Muon System is a general purpose detector like ATLAS . It's 21m long,15m in diameter and weighs 14000 tonne. It was constructed in 15 sections at ground level before being lowered . The Emphasis is on muon detection. Muons general arise from decay of massive unstable(possibly interesting) particles . The material between detector and IP virtually absorbs other particles. Maximum energy reached is similar to ATLAS. It was also involved in detection of higgs.

3.1 CMS configuration

3.1.1 Magnet system

It has a Large single superconducting solenoid. The charged particles bend in transverse plane contrast to ATLAS. It's dimensions are 13m in length and inner diameter of 5.9m . It Creates a magnetic field of 4T. It operates at Liquid He temperature 4K so it's inside a vacuum vessel(2 concentric cylinders) isolated from ex-

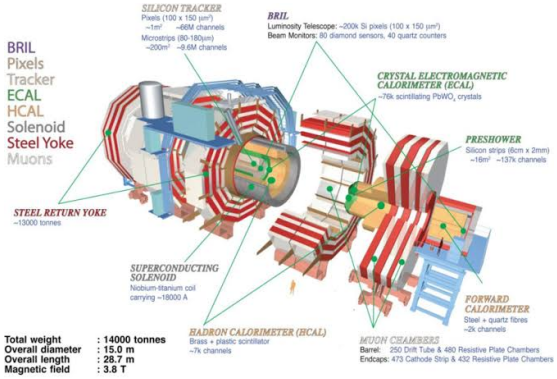


Figure 6: CMS

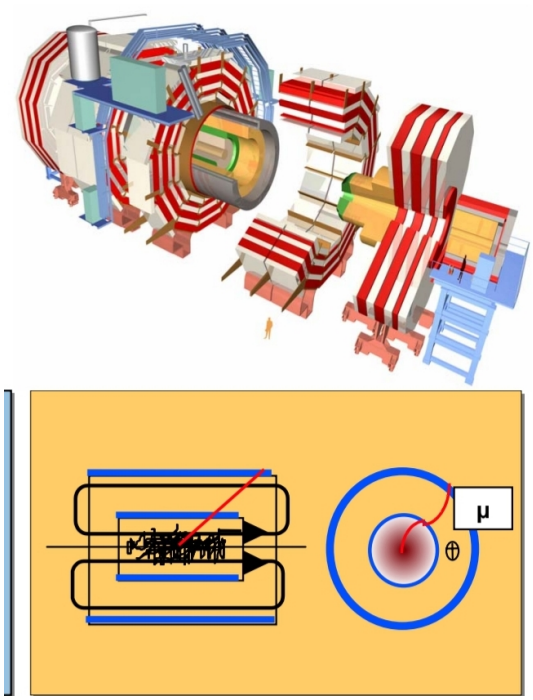


Figure 7: CMS Magnet system

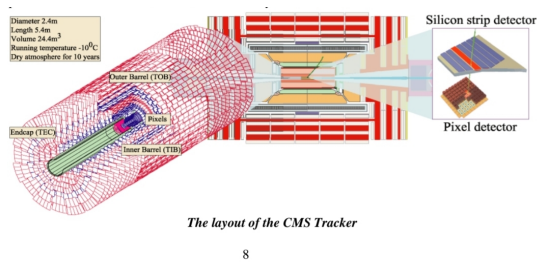


Figure 8: CMS Tracker system

terior.It consists of a return yoke with magnetic field 2T to control magnetic field outside solenoid. Return yoke acts as the main support system of CMS .

3.1.2 Tracker system

It contains concentric layers of Silicon sensors. It measures momenta of charged particles and reconstructs interaction vertices. The formula used is $p = R \times 0.3B$ where R is in m, p in GeV/c^2 and B in T. Consists of Si pixel and microstrip layers in cylindrical volume . Large volume is required for significant bending of high energy particles. Pixel detectors are close to IP while Si microstrip covers Large volume for momentum measurement. The concentric layers are thin so that natural trajectories are not diverted.

3.1.3 Electromagnetic Calorimeter

ECAL measures energy and incidence position of electron/positron and photon. Main part is Constructed from truncated pyramid crystals of lead tungsten ($PbWO_4$) . It has a high effective Z leading to small radiation length.The amount of light created is linear function of

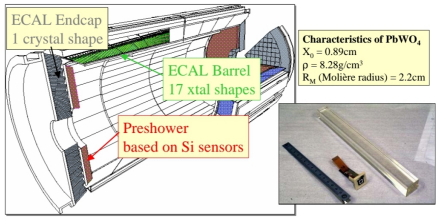


Figure 9: CMS ECAL

particle energy incident on Scintillator. Its snalogous to ATLAS LAr Calorimeter.

3.1.4 Hadronic Calorimeter

It surrounds ECAL and plays an important role in identification of quarks,gluons and neutrino .Its divided into 3 regions(barrel HB,endcap HE and forward HF) . HE and HB are sampling Calorimeter . The Scintillators in them emit blue-violet light proportional to energy of particle. HF calorimeters are present at two extreme ends of CMS.The energy of jets is measured from the Cherenkov light produced as charged particles traverse the quartz fibres .

3.1.5 Muon Chamber

CMS is significantly optimized for detection and measurement of muons. Drift tubes are located outside of solenoid in barrel region (passing muon ionizes the gas in it). Cathode strip chambers are present in endcap region. Perpendicular strips of wire in it help in processing precise xy coordinate . It also uses RPC like ATLAS.

3.1.6 Trigger and Data Acquisition System

To manage storage the 40MHz input rate must be lowered to 100Hz. At L1(hardware level) the data is temporarily held for $3\mu s$. L1 identifies high energy signature. It reduces from 40MHz to 100kHz. HLT(software level) runs on about 1000 commercial computer. It can handle 1TB/s of data. Finally it's filtered to 100Hz . DAQ-
(1) Readout: Takes Data accepted by L1,
(2) Selection: Selects physically interesting Data,
(3)Archiving and monitoring .

3.1.7 CMS physics experiment

(1)In 2010(1st run) 7TeV of pp collisions and first Pb ion collisions were observed.
(2)In 2012 8TeV pp collisions took place. It detected Higgs and a neutral excited xi-b baryon .
(3)Beginning of 2nd run (2015)the energy increased to 13TeV. In 2018 Higgs decaying to bottom quark pair was observed.
(4)The run 3 began in 2022 with COM energy going 13.6TeV.
(5)The top quark has long been thought to decay too quickly to form a bound state . Recent Data from CMS suggest that top forms a bou d state with anti top - toponium. The relative velocity was used to detect this rather than direct mass measure .
(6)Last year it was reported that CMS measured the properties of family of three all charm tetra quarks. They analyzed the J/Psi decay and further muon decay .

(7)In 2025 CMS gave it's first W boson mass 80360.2 MeV with uncertainty of 9.9 MeV which is the most precise .
(8)In 2024 $\gamma\gamma$ to 2 tau was observed for first time . This also gave the most precise anomalous magnetic moment of tau .
(9)In 2024 CMS reported that with the help of AI they were able to find first evidence of CP violation in decay of strange beauty meson into muons and kaons.

4 Acknowledgement

I would like to acknowledge my friend and co-partner in this project Pushkar Tripathi (CHEP 2025-) .I would also like to acknowledge my Professor Dr. Balasubramanian Ananthnarayan for giving me this opportunity. I would also like to acknowledge my Family and Friends who all helped in their possible ways.

5 References

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